Raagul Sundaralingam

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WORK EXPERIENCE

CapTech Arlington, VA
Software Engineering Consultant *July 2025 - Present*

- Co-built a config-driven PySpark ETL platform on AWS EMR for a top-5 U.S. financial institution, cutting pipeline runtimes from 12+ hrs to 20–30 min with built-in data-governance checks; adopted for additional lines of business.
- Owned the custom-SQL layer and, as sole owner of historical data copy, built a config-driven tool that migrated 10 GB–20 TB datasets in ≤1/3 the time of prior 24+ hr methods, then handed it off to the client for future migrations.
- Built an LLM app (GPT-4 on Azure, Next.js, FastAPI, Docker) that bulk-converts new-hire résumés into CapTech's Workday format, replacing a manual process; an LLM-judge eval loop over hundreds of résumés raised accuracy from ~60% to near-error-free.
- Built an LLM error-diagnostics feature for CapTech's firm-wide internal deployment platform (terminal-style Next.js UI, GPT-4 explanations and fix suggestions for failed deploys); demoed to the platform team and helped integrate it into production.
- Built an agentic certification-study app (Next.js, FastAPI, Ollama): a ReAct-loop agent with custom MCP-style tools researches documentation, models learner knowledge, and generates study guides, flashcards, and practice exams.

Machine Analysis of Political Speech Lab, Ohio State University Columbus, OH
Research Assistant *January 2024 - June 2025*

- Architected a Python multimodal pipeline on Linux that automates ingestion and processing of large-scale political datasets.
- Automated extraction with WhisperX and Llama-3 into a SQL database, cutting 1.5-hr video processing to ~13 min and producing queryable datasets on presidential debate candidates.

EXTRACURRICULARS AND LEADERSHIP

Mimir - AI-powered Academic Assistant *January 2025 - April 2025*
Project Lead

- Led development of a RAG system (Weaviate, Ollama, FastAPI) over thousands of embedded ArXiv papers, delivering citation-grounded suggestions in a real-time Next.js editor.

Acquired.Beauty - Makeup Shopping Platform *January 2025 - April 2025*
Project Lead

- Led a 4-person team building a Next.js/FastAPI/Supabase marketplace with a Firecrawl + Claude API pipeline to enrich product data and a matching algorithm for personalized recommendations; user research shaped the roadmap.

EDUCATION

The Ohio State University Columbus, OH
Bachelor of Science in Computer Science & Engineering *May 2025*
Specialization: Artificial Intelligence *GPA: 3.7*
Awards: Dean's List; 1st Place ShowOHI/O 2023; 2nd Place HackAI 2023; TartanHacks 2025 MakingWaves Track Winner
Certifications: AWS Certified Solutions Architect – Associate (SAA-C03) [August 2026]

SKILLS

Programming Languages: Python, TypeScript/JavaScript, SQL, Java, C/C++
AI/LLM: RAG, agents and tool use (ReAct, MCP-style), LLM evals, Claude API, Ollama, PyTorch, HuggingFace, Weaviate, Qdrant
Web/Backend: Next.js, React, React Native, FastAPI, Node.js, PostgreSQL, Supabase
Cloud/Data: AWS, PySpark, Snowflake, Docker, Kubernetes, GCP, GitHub Actions